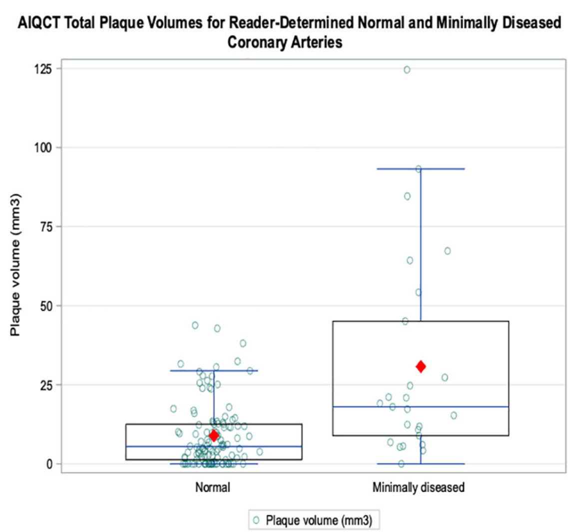
identified 96/116 (82.8%) of R_{NORMAL} cases and $\geq$ 15.3mm^3 identified 15/25 (60%) of $R_{MINIMAL}$ cases (Figure).

Conclusions: AIQCT finds substantially higher total plaque volume in reader-determined minimally diseased coronary arteries compared to normal coronary arteries, demonstrating promising compatibility with reader interpretation. Our exploratory analysis suggests AIQCT total plaque volume of < 15.3 mm^3 may be consistent with reader-determined normal coronary arteries.



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MACHINE LEARNING FOR PREDICTING LONG-TERM MAJOR ADVERSE CARDIOVASCULAR EVENTS BASED ON CLINICAL RISK FACTORS AND CORONARY ARTERY CALCIUM IMAGING METRICS: RESULTS FROM THE EISNER STUDY

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Introduction: The coronary artery calcium (CAC) score from a non-contrast ECG-gated cardiac CT is a well-established tool for assessing cardiovascular risk and guiding preventive therapy decisions. Research has shown that incorporating additional metrics from the CAC scan, such as the regional distribution and extent of CAC and CAC density, may improve the accuracy of risk prediction. We sought to evaluate the prognostic value of a machine learning (ML) model that incorporates both clinical risk factors and multiple CAC imaging metrics to predict major adverse cardiac events (MACE).

Methods: This study included 2,032 subjects (age: 55.6±9, 59% male) who underwent a CAC scan and were followed up for 14 years for MACE, which included myocardial infarction, late revascularization, and cardiac death. In addition to the conventional Agatston CAC score, we analyzed several other CAC metrics, including total and density-based CAC volumes [assessed by CAC volume (mm3) according to CAC Hounsfield unit], the number of calcified plaques, proximal involvement, and aortic calcium. An extreme gradient boosted tree model (XGBoost) was trained and validated using 10-fold cross validation to predict MACE. The performance of the ML including all variables (clinical risk factors, CAC score, and additional CAC metrics) and the conventional clinical approaches using Atherosclerotic Cardiovascular Disease (ASCVD) risk and CAC scores were compared using receiver-operating characteristic curve analysis. Feature importance results from XGBoost were analyzed to identify the most important predictors of MACE.

Results: Over a median follow-up of 14.4 years (interquartile range: 13.4-15.8), 206 patients (10%) experienced MACE. The ML model outperformed the combined ASCVD risk and CAC scores, achieving a significantly higher area under the curve [0.78 (95% confidence interval (CI): 0.73-0.83) vs. 0.75 (95% CI: 0.70-0.81), p<0.01; Figure A]. Feature importance analysis revealed that the ML algorithm identified several additive CAC metrics as predictors of MACE, including low and intermediate density calcium volume, proximal CAC score, extent of CAC score, and number of plaques (Figure B).

Conclusions: ML incorporating both clinical and multiple CAC imaging metrics significantly improved the prediction of MACE compared to the traditional approach using ASCVD risk and CAC scores. Additive CAC metrics are important features in enhancing risk stratification for predicting MACE.

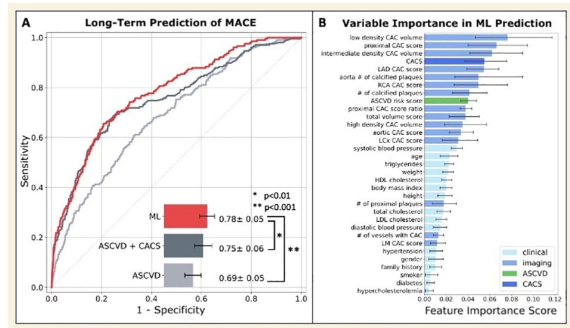

Figure A Receiver-operating characteristic curves for the 14-year prediction of MACE (206/2032 subjects). The ML algorithm outperformed the ASCVD risk score (*p*<0.001) and the combined ASCVD risk and CAC score (*p*<0.01) with statistical significance as determined by the DeLong Test. **Figure B** The ML algorithm identified low and intermediate density calcium volume, proximal CAC score, extent of CAC score (number of vessel involvement) and number of plaques as the most important predictors of MACE in feature importance analysis.

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STREAMLINED CAROTID ARTERY CALCIFICATION LABELING FOR CTA SCANS

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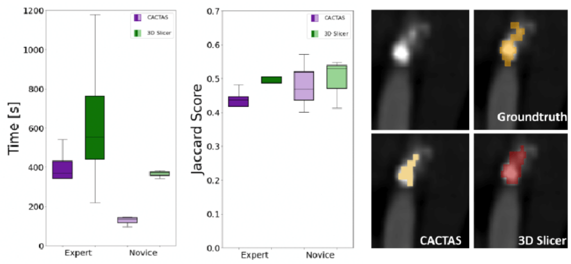
Introduction: We require vast amounts of manually labeled calcifications to train advanced machine learning classifiers for carotid artery plaque detection and phenotyping high-risk calcifications for ischemic stroke patients. We developed CTOOL, a web-based software for streamlined calcium labeling with minimal user interaction.

Methods: We designed and implemented CTOOL with radiologists to allow single-click plaque annotation. We perform 3D region growing based on a user-selected CT Hounsfield Unit (HU) intensity with a configurable tolerance threshold to include neighboring voxels. Visualization was set to window/level of 130/1500 HU. We selected 64 retrospective neck CTA dataset of ischemic stroke from the Penn Stroke Registry. For expert validation, we randomly selected 7 CTA scans from this data collection and divided them into a training set (N=2) and a testing set (N=5). A radiologist performed calcification annotations with CTOOL vs. manual segmentation with 3D Slicer. We time the experiment and measure accuracy as Jaccard index against fully-manual annotations with a multiple radiologist-consensus across a 4cm segment of the carotid bifurcation. We then recruited N=6 participants (avg. age 30.67yrs, 4 female, IRB approved) for a between-subjects user study without prior experience with either annotation software to compare speed across observers. We randomly assigned the participants and asked them to annotate carotid plaque in a single CTA scan after a brief training period of 5 minutes.

Results: Plaque annotations with CTOOL were faster than with 3D Slicer (expert w/ CTOOL 376.6±113.52s vs 631.2±325.88s, novice w/ CTOOL 126.59±21.57s vs 366.09±16.77s, t4=-12.40, p<0.0001). Annotation accuracy between both softwares was comparable (expert Jaccard w/ CTOOL 0.537±0.077 vs 0.464±0.238, novice Jaccard w/ CTOOL 0.481±0.070 vs 0.496±0.060). However, novice annotations will require expert validation or proofreading. All participants completed the NASA-TLX questionnaire to reveal lower mental, physical, and temporal demand with our software (expert w/ CTOOL 12.67 vs 16.334, novice w/ CTOOL 28.33±30.880 vs 33.33±14.727).

Conclusions: Our streamlined plaque labeling software allows 2.89x faster annotations (1.68x for experts) with comparable accuracy to manual segmentation.



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COMPARISON OF AI-BASED CORONARY CTA INTERPRETATION AND QCA FOR CORONARY STENOSIS EVALUATION

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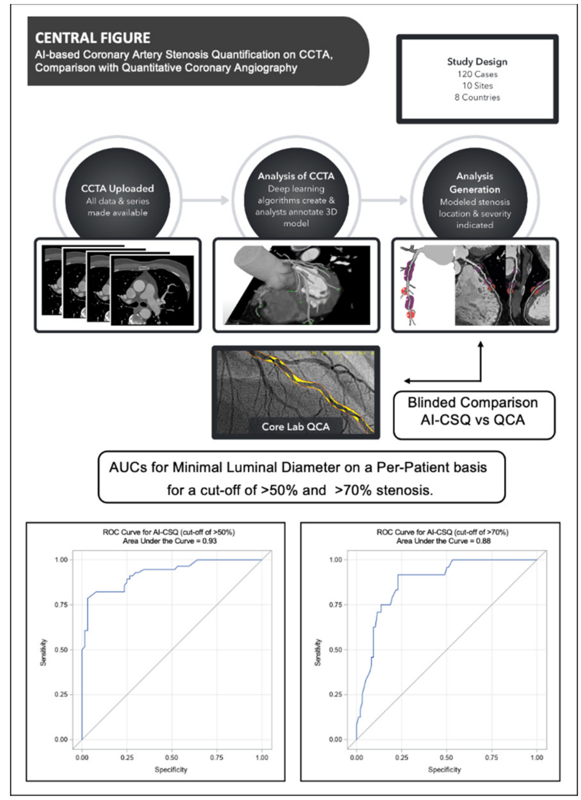
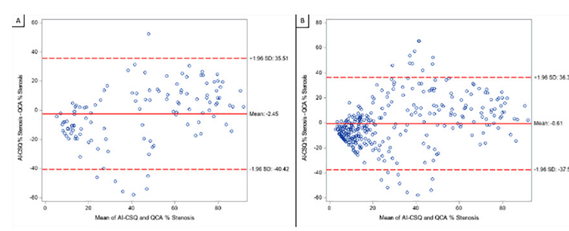
Introduction: Coronary CT angiography (CCTA) has become an established tool in the diagnostic work-up of patients with suspected coronary artery disease (CAD). Despite the high diagnostic performance, clinical CCTA interpretation has variability between readers with different degrees of correlation to invasive coronary angiography (ICA). AI-workflows could provide context to better interpret CCTA and overcome these limitations. We sought to evaluate the performance of a new AI-based tool by comparing the quantified stenosis severity with a gold-standard reference derived from invasive quantitative coronary angiography (QCA).

Methods: A total of 120 patients were included in the analysis; mean age was 59.7 ±10.8 years and 47 (39.2%) were female. All patients underwent CCTA, invasive coronary angiography and QCA. Quantitative analysis of CAD stenosis severity was performed using an AI-based Coronary Stenosis Quantification (AI-CSQ) software service. Blinded comparison between QCA and AI-CSQ was measured on a per-patient and per-vessel basis.

Results: The per-patient AI-QCS diagnostic sensitivity, specificity, accuracy, positive predictive value, and negative predictive value was 82%, 84%, 83%, 82%, and 84%, respectively, for minimal luminal diameter (MLD) stenosis >50%; and 75%, 85%, 83%, 56%, and 93%, respectively, for MLD stenosis >70%. The AUC in ROC analysis on a per-patient and per-vessel basis was 0.93 and 0.92 respectively at a ≥50% cut-off, and 0.88 and 0.93 respectively at a ≥70% cut-off.

Conclusions: AI-CSQ for quantitative CCTA stenosis severity evaluation displays a high diagnostic performance compared to QCA both on a per-patient and per-vessel basis with high sensitivity for stenosis detection. Further studies are needed to better understand how this AI-based approach can be integrated into clinical decision-making in a fashion that facilitates faster, more reproducible, and accurate clinical CCTA interpretation.





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EVALUATION FOR ARTIFICIAL INTELLIGENCE-BASED CAC SCORING MODEL EFFICIENCY AND ACCURACY

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